

EMPLOYEE

Java OOP Program Documentation

1. Question

Write a Java program to demonstrate constructor overloading using a worker class with different constructors to initialize employee name, ID, designation, salary, and promotion status. Display the details of all worker objects.

2. Execution Steps

1. Open Eclipse IDE or a terminal.
2. Create a Java project/package named Excercise.
3. Create a Java class named worker.
4. Copy the complete code into Employee.java.
5. Make sure that public class name is Employee and file name is Employee.java.
6. Declare the worker details: name, id, designation, salary, and promotion status.
7. Create a default constructor worker() and assign default values.
8. Create a three-parameter constructor to initialize name, ID, and designation.
9. Create a four-parameter constructor to initialize name, ID, designation, and salary.
10. Create a five-parameter constructor to initialize name, ID, designation, salary, and promotion status.
11. Create a display() method to print all the worker details.
12. Create four worker objects using the four constructors.
13. Call the display() method for each object.
14. Display the details of all four workers.
15. Run the Employee class containing the main() method.
16. Stop the program.

3. Algorithm

Step 1: Start.

Step 2: Define a class named worker.

Step 3: Declare the variables name, id, designation, salary, and promotion status.

Step 4: Define a default constructor with no parameters and initialize the variables with default values.

Step 5: Define a constructor with three parameters: name, id, and designation. Initialize these values and set salary to 0.0 and promotion status to false.

Step 6: Define a constructor with four parameters: name, id, designation, and salary. Initialize these values and set promotion status to false.

Step 7: Define a constructor with five parameters: name, id, designation, salary, and promotion status. Initialize all the values.

Step 8: Define a display() method to print the details of a worker.

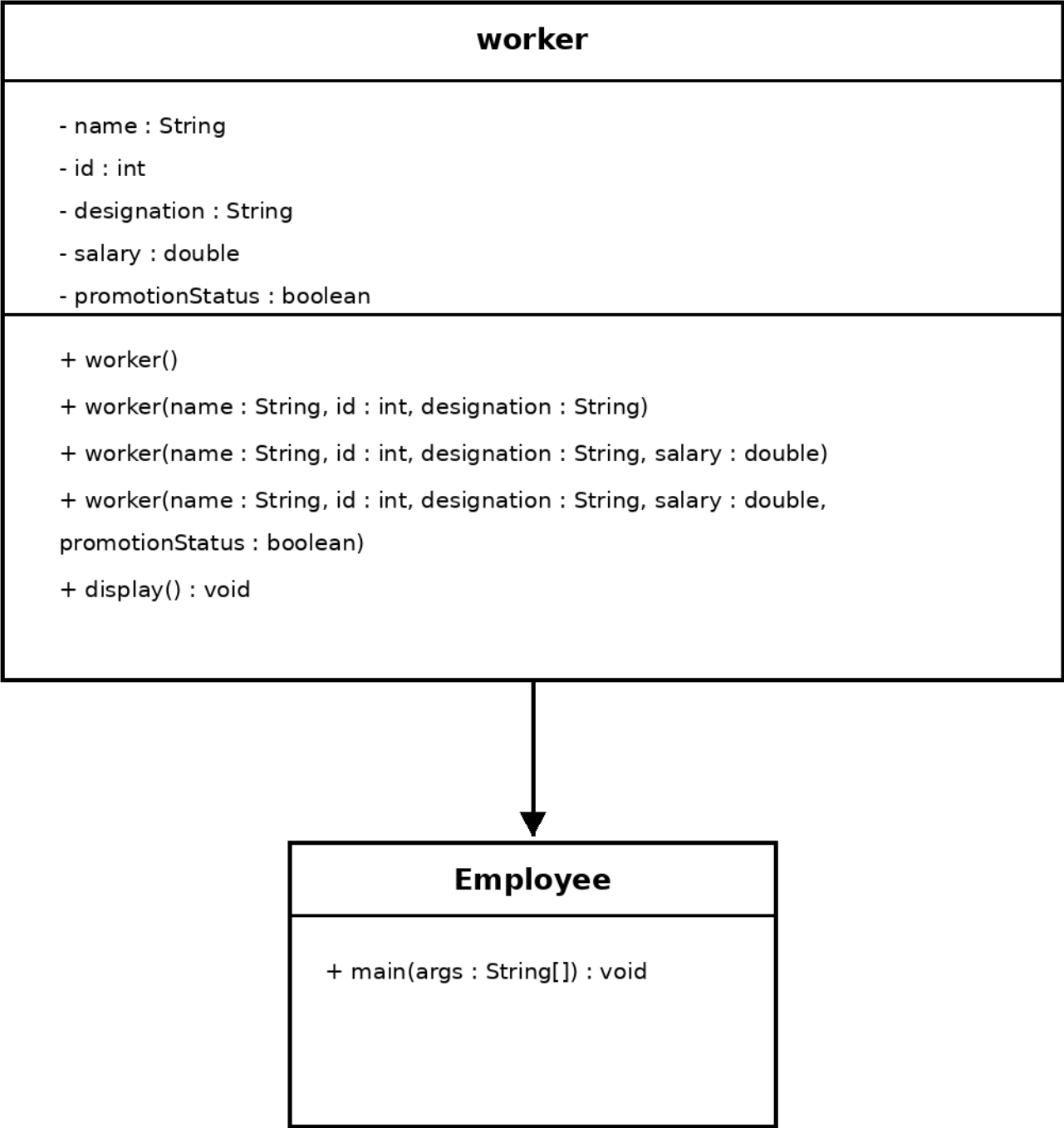
Step 9: In the main() method, create four objects: w1 using the default constructor, w2 using the three-parameter constructor, w3 using the four-parameter constructor, and w4 using the five-parameter constructor.

Step 10: Call display() for w1, w2, w3, and w4.

Step 11: Display all worker details.

Step 12: Stop.

4. UML Diagram



5. Java Source Code

```
package Excercise;
class worker{
    String name;
    int id;
    String designation;
    double salary;
    boolean promotionStatus;

    worker() {
        this.name = "Not Assigned";
        this.id = 0;
        this.designation = "Not Assigned";
        this.salary = 0.0;
        this.promotionStatus = false;
    }

    worker(String name, int id, String designation) {
        this.name = name;
        this.id = id;
        this.designation = designation;
        this.salary = 0.0;
        this.promotionStatus = false;
    }

    worker(String name, int id, String designation, double salary) {
        this.name = name;
        this.id = id;
        this.designation = designation;
        this.salary = salary;
        this.promotionStatus = false;
    }

    worker(String name, int id, String designation, double salary, boolean promotionStatus) {
        this.name = name;
        this.id = id;
        this.designation = designation;
        this.salary = salary;
        this.promotionStatus = promotionStatus;
    }

    void display() {
        System.out.println("Employee Name : " + name);
        System.out.println("Employee ID : " + id);
        System.out.println("Designation : " + designation);
        System.out.println("Salary : " + salary);
        System.out.println("Promotion Status : " + promotionStatus);
        System.out.println("-----");
    }
}

public class Employee {
    public static void main(String[] args) {
        worker w1 = new worker();
        worker w2 = new worker("Rahul", 101, "Software Engineer");
        worker w3 = new worker("Priya", 102, "Software Engineer", 50000);
        worker w4 = new worker("Arun", 103, "Manager", 80000, true);
        w1.display();
        w2.display();
        w3.display();
        w4.display();
    }
}
```

6. Output – Terminal Screenshot

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22631.3593]
(c) Microsoft Corporation. All rights reserved.

C:\Users\abc\Desktop\Excercise>java Excercise.Employe
Employee Name      : Not Assigned
Employee ID       : 0
Designation       : Not Assigned
Salary           : 0.0
Promotion Status  : false
-----
Employee Name      : Rahul
Employee ID       : 101
Designation       : Software Engineer
Salary           : 0.0
Promotion Status  : false
-----
Employee Name      : Priya
Employee ID       : 102
Designation       : Software Engineer
Salary           : 50000.0
Promotion Status  : false
-----
Employee Name      : Arun
Employee ID       : 103
Designation       : Manager
Salary           : 80000.0
Promotion Status  : true
-----
C:\Users\abc\Desktop\Excercise>_
```